

# Models of Curves and Valuations

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# Preface

This is a preliminary version of an open book developed as part of the MCLF project.

The text is work in progress. Individual versions should be cited by referring to a specific Git commit or release.

# Chapter 1

## Valued Fields and Models

### 1.1 Models of Curves

#### 1.1.1 Basic Definitions

Let  $K$  be a field endowed with an additive valuation of rank 1,  $v_K : K \rightarrow \hat{\mathbb{R}} := \mathbb{R} \cup \{\infty\}$ . Let  $\Gamma_K := v_K(K^*) \subset \mathbb{R}$  be the value group of  $v_K$ ,  $\mathcal{O}_K \subset K$  its valuation ring with maximal ideal  $\mathfrak{m}_K$  and  $k := \mathcal{O}_K/\mathfrak{m}_K$  its residue field. We assume that  $v_K$  is discrete,  $K$  is complete with respect to  $v_K$  and  $k$  is perfect. Let  $X$  be a smooth, projective and absolutely irreducible curve over  $K$ .

**Definition 1.1.** A *model*  $\mathcal{X}$  of  $X$  is a proper, flat and normal  $\mathcal{O}_K$ -scheme with an identification of its generic fibre  $\mathcal{X}_g$  with  $X$ .

**Example 1.2.** Let  $K = \mathbb{Q}_p$ ,  $\mathcal{O}_K = \mathbb{Z}_p$  and  $X = \mathbb{P}^1(K)$ , then  $\mathcal{X} = \mathbb{P}^1(\mathcal{O}_K)$  is a model of  $X$  with the evident isomorphism from  $\mathcal{X}_g = \mathcal{X} \times_{\mathcal{O}_K} K$  to  $\mathbb{P}^1(K)$ . The special fibre<sup>1</sup>  $\mathcal{X}_s$  has one irreducible component which is isomorphic to  $\mathbb{P}^1(k)$ .

*Remark 1.3.* The condition of a model being flat is equivalent to it being integral: If  $\mathcal{X} \rightarrow \operatorname{Spec} \mathcal{O}_K$  is an integral scheme, then for every  $x \in \mathcal{X}$ , the stalk  $\mathcal{O}_{\mathcal{X},x}$  is an integral ring extension of  $\mathcal{O}_K$ . Thus  $\mathcal{O}_{\mathcal{X},x}$  is a torsion-free module over the discrete valuation ring  $\mathcal{O}_K$ , hence flat by [?, Corollary 1.2.14]. Conversely, the integrality of a model  $\mathcal{X}$  of  $X$  follows from its flatness and the integrality of  $X$  by [?, Proposition 4.3.8].

*Remark 1.4.* A point  $x \in \mathcal{X}$  of codimension 1 which is regular is normal. This is conversely true by Serre's criterion (see [?, Lemma 8.2.21]).

**Example 1.5.** Consider the curve  $X$  defined by  $y^2 = x^3 + px$  over  $\mathbb{Q}_p$  with  $p \neq 2$ . Then

$$\mathcal{X} := \operatorname{Proj} \mathbb{Z}_p[X, Y, Z]/(Y^2Z - X^3 - pXZ^2) \quad (1.1)$$

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<sup>1</sup>That is, the fibre of the closed point of  $\operatorname{Spec} \mathcal{O}_K$ .

is a model of  $X$ . It is clearly a proper scheme over  $\mathcal{O}_K = \mathbb{Z}_p$ . For the flatness it suffices to show that  $R := \mathbb{Z}_p[X, Y, Z]/(Y^2Z - X^3 - pXZ^2)$  is a torsion-free  $\mathbb{Z}_p$ -module by Remark 1.3. But this is true since  $Y^2Z - X^3 - pXZ^2$  is irreducible in  $\mathbb{Z}_p[X, Y, Z]$ .

**TBD: Normality**

### 1.1.2 Valuations of Type II

In this section, we assume the valuation  $v_K$  to be *normalised*, i.e.  $v_K(K^*) = \mathbb{Z}$ .

**Definition 1.6.** Let  $F_X$  be a function field of  $X$  over  $K$ . A valuation  $v : F_X \rightarrow \hat{\mathbb{R}}$  is said to be *of type II*, if

1.  $v|_K = v_K$ ,
2.  $v$  is discrete, and
3. the extension of residue fields  $k(v)|k$  has transcendence degree 1.

A valuation of this type is also known as *geometric valuation*. We write  $V_{\text{II}}(F_X)$  for the set of all valuations of type II of  $F_X$ . The number

$$m(v) := [\Gamma_v : \Gamma_K] \quad (1.2)$$

is called the *multiplicity of  $v$* , where  $\Gamma_v := v(F_X^*)$  is the value group. We say that  $v$  is *weakly unramified*, if  $m(v) = 1$ .

**Example 1.7.** The *Gauß valuation* on the polynomial ring  $K[x]$  is defined as

$$v_g : K[x] \rightarrow \hat{\mathbb{Q}}, \quad v_g \left( \sum_i a_i x^i \right) = \min_i (v_K(a_i)). \quad (1.3)$$

Since  $k(v_g) = k(\bar{x})$  (where  $\bar{x}$  is the image of  $x$  in  $k(v_g)$ ), the Gauß valuation extends naturally and uniquely to  $K(x)$ , meaning that we have  $v_g \in V_{\text{II}}(K(x))$ . We will later see that  $\{v_g\}$  defines the model  $\mathcal{X} = \mathbb{P}^1(\mathbb{Z}_p)$  of  $\mathbb{P}^1(\mathbb{Q}_p)$ .

Let  $\mathcal{X}$  be a model of  $X$ . We wish to assign a valuation in  $V_{\text{II}}(F_X)$  to each generic point of the special fibre  $\mathcal{X}_s$ . Suppose that  $Z$  is an irreducible component of  $\mathcal{X}_s$  with generic point  $\xi$ . Since  $\xi$  has codimension 1 in  $\mathcal{X}$ , the stalk  $\mathcal{O}_{\mathcal{X}, \xi}$  is a discrete valuation ring. Thus, we can define a valuation on  $F_X$ ,

$$v_Z : \text{Frac}(\mathcal{O}_{\mathcal{X}, \xi}) = F_X \rightarrow \hat{\mathbb{Q}}. \quad (1.4)$$

We set

$$V(\mathcal{X}) := \{v_Z : Z \subset \mathcal{X}_s \text{ is an irreducible component}\}. \quad (1.5)$$

**Proposition 1.8.** *The valuation defined in 1.4 is of type II.*

*Proof.* Since  $\mathcal{X}$  is a flat  $\mathcal{O}_K$ -scheme  $\mathcal{O}_{\mathcal{X},\xi}$  dominates  $\mathcal{O}_K$ , and we may scale  $v_Z$  such that the restriction of  $v_Z$  to  $K$  is  $v_K$ . Furthermore, we have a closed immersion  $Z \hookrightarrow \mathcal{X}$  by definition, inducing a surjection  $\mathcal{O}_{\mathcal{X},\xi} \rightarrow \mathcal{O}_{Z,\xi}$ . Thus  $\text{trdeg}(k(v_Z)) = \text{codim}(\mathcal{O}_{Z,\xi}) = \text{codim}(v_Z) = 1$ , showing  $v_Z \in V_{\text{II}}(F_X)$ .  $\square$

**Definition 1.9.** The *multiplicity* of the irreducible component  $Z$  of  $\mathcal{X}_s$  containing the generic point  $\xi$  is defined as

$$m(Z) := \text{length}(\mathcal{O}_{\mathcal{X},\xi}). \quad (1.6)$$

**Proposition 1.10.** *It holds  $m(v_Z) = m(Z)$ .*

*Proof.* As the model  $\mathcal{X}$  is normal, the stalk  $\mathcal{O}_{\mathcal{X},\xi}$  is a discrete valuation ring. Let  $\pi_\xi$  denote an uniformizing parameter, then

$$\mathcal{O}_{\mathcal{X}_s,\xi} = \mathcal{O}_{\mathcal{X},\xi}/(\pi_\xi). \quad (1.7)$$

Thus, the length of the stalk at  $\xi$  coincides with the length of the chain of ideals generated by the uniformizing parameter of the quotient ring: Since  $(\pi_\xi) = (\pi_{Z,\xi}^m)$ , where  $m = (\Gamma_{v_Z} : \Gamma_K)$ , it holds  $m(v_Z) = \text{length}(\mathcal{O}_{\mathcal{X},\xi})$  as desired.  $\square$

**Definition 1.11.** A *modification* of a model of  $X$  is a morphism  $\mathcal{X}' \rightarrow \mathcal{X}$  of  $\mathcal{O}_K$ -schemes which induces the identity on the generic fibre.

*Remark 1.12.* Since a model is proper over  $\mathcal{O}_K$ , a modification is a proper morphism. Thus since it is dominant, it is also surjective.

The main goal of the first two sections of this chapter is to establish the following theorem.

**Theorem 1.13.** *The map*

$$\mathcal{X} \mapsto V(\mathcal{X}). \quad (1.8)$$

*defines a monotone bijection between*

- *the set of isomorphism classes of models of  $X$ , and*
- *the set of finite, nonempty subsets of  $V_{\text{II}}(F_X)$ .*

In the remainder of this section, we prove injectivity of this map.

**Lemma 1.14** ([?, Lemma 3.6]). *Let  $\mathcal{X}$  be a model of  $X$  and let  $\mathcal{X}'$  be a model of another irreducible smooth projective curve  $X'$ . Let  $\varphi : \mathcal{X}' \rightarrow \mathcal{X}$  be a morphism of  $\mathcal{O}_K$ -schemes which induces a cover  $X' \rightarrow X$  on the generic fibre. Let  $\eta$  be a generic point of  $\mathcal{X}_s$ , then the preimage of  $\eta$  under  $\varphi$  is non-empty and consists of generic points of  $\mathcal{X}'_s$ . If  $\varphi$  is birational, then the preimage consists of precisely one generic point.*

*Proof.* Let  $Z \subset \mathcal{X}_s$  be the irreducible component that contains  $\eta$ . Since  $\varphi$  is surjective (cf. Remark 1.12), there is a point  $\eta'$  on an irreducible component  $Z' \subset \mathcal{X}'_s$  which is mapped to  $\eta$ . Since  $\varphi$  is dominant, the induced ring homomorphism  $\mathcal{O}_{\mathcal{X},\eta} \rightarrow \mathcal{O}_{\mathcal{X}',\eta'}$  is injective. This is an extension of discrete valuation rings and thus  $\eta' \in Z'$  must be generic. In particular, if  $\varphi$  is birational, then both rings are integrally closed in the same field of fractions which makes them equal. Therefore,  $\eta'$  must be unique since  $\mathcal{X}'$  is separated.  $\square$

**Lemma 1.15** ([?, Lemma 3.7]). *Let  $\varphi : \mathcal{X}' \rightarrow \mathcal{X}$  be a modification of models of  $X$ . Then  $V(\mathcal{X}) \subset V(\mathcal{X}')$ .*

*Proof.* This is a consequence of Proposition 1.8 and Lemma 1.14.  $\square$

We have a partial order of models of  $X$  given by  $\mathcal{X} \leq \mathcal{X}'$  if there exists a modification  $\mathcal{X}' \rightarrow \mathcal{X}$ . By the previous lemma, this gives a partial order on the corresponding finite subsets of  $V_{\text{II}}(F_X)$  via inclusion maps.

**Lemma 1.16** ([?, Lemma 3.8]). *Let  $\mathcal{X}$  and  $\mathcal{X}'$  be models of  $X$  with  $V(\mathcal{X}) \subset V(\mathcal{X}')$ . Then there is a modification  $\mathcal{X}' \rightarrow \mathcal{X}$ .*

*Sketch of the proof.* TBD.  $\square$

**Proposition 1.17** ([?, Proposition 3.9], [?, Corollary 8.3.23]). *Two models  $\mathcal{X}$  and  $\mathcal{X}'$  of  $X$  are isomorphic as models of  $X$  if and only if  $V(\mathcal{X}) = V(\mathcal{X}')$ . In particular, the map defined in (1.8) is injective.*

*Proof.* Suppose that  $V(\mathcal{X}) = V(\mathcal{X}')$ . Then there exist morphisms  $\mathcal{X} \rightarrow \mathcal{X}'$  and  $\mathcal{X}' \rightarrow \mathcal{X}$ . The composition of these morphisms belongs to the equivalence class of the identity morphism, when considered as a rational map.  $\square$

**Example 1.18.** We continue with Example 1.5 of the model

$$\mathcal{X} = \text{Proj } \mathbb{Z}_p[X, Y, Z]/(Y^2Z - X^3 - pXZ^2)$$

of the elliptic curve  $X : y^2 = x^3 + px$  defined over  $\mathbb{Q}_p$ . We have already checked that it is a model. Let  $Z$  be the irreducible component in  $\mathcal{X}_s$  containing the generic point  $\xi$  and denote the corresponding valuation by  $v_Z$ .  $Z$  is defined by  $\bar{X}_p : y^2 = x^3$ , thus  $Z = \mathcal{X}_s$  and  $V(\mathcal{X}) = \{v_Z\}$ .

We want to describe  $v_Z$  as

$$v_Z|_{F_X} = v_g \quad \text{Why do we need to restrict?}$$

with respect to the  $x$ -coordinate. Using Lemma (In the third talk), we see that  $v_Z(p) = 1$  and  $v_Z(x) = 0$ , since the horizontal divisor on  $x = 0$  intersects  $\bar{X}_p$  at a unique point

## 1.2 The Proof of Theorem 1.13

For the proof of Theorem 1.13, it remains to show the surjectivity of the map ((1.8)), i.e., for a given finite, non-empty set of valuations  $V \subset V_{\text{II}}(F_X)$ , there is a model  $\mathcal{X}$  of  $X$  such that  $V = V(\mathcal{X})$ . We will prove this in two steps, which we state in the following proposition.

**Proposition 1.19** ([?, 3.14, 3.16], [?, 8.3.36]). *(i) Given model  $\mathcal{X}$  and a valuation  $v \in V_{\text{II}}(F_X)$ , there exists a modification  $\varphi: \mathcal{X}' \rightarrow \mathcal{X}$  such that  $v \in V(\mathcal{X}')$ .*

*(ii) Given a model  $\mathcal{X}$  and a nonempty subset  $V \subset V(\mathcal{X})$ , there exists a modification  $\varphi: \mathcal{X} \rightarrow \mathcal{X}'$  such that  $V = V(\mathcal{X}')$ .*

After proving Proposition 1.19, the proof is complete: Starting with a model  $\mathcal{X}$  of  $X$ , we can apply part (i) for every  $v \in V \subset V_{\text{II}}(F_X)$  to obtain a new model that induces all valuations in  $V$ , but possibly more. We then apply part (ii) to construct a model that induces exactly the valuations in  $V$ .

### 1.2.1 Proof of the First Part

We fix a model  $\mathcal{X}$  of  $X$  and a type II valuation  $v \in V_{\text{II}}(F_X)$ .

**Definition 1.20.** A point  $x \in \mathcal{X}$  is called a *centre* of  $v$  if there exists a morphism  $f: \text{Spec } \mathcal{O}_v \rightarrow \mathcal{X}$  of  $\mathcal{O}_K$ -schemes which maps the closed point to  $x$  and makes the following diagram commute:

$$\begin{array}{ccc} \text{Spec } \mathcal{O}_v & \xrightarrow{f} & \mathcal{X} \\ & \nwarrow \quad \nearrow & \\ & \text{Spec } F_X & \end{array}$$

Here, the left map is induced by the inclusion  $\mathcal{O}_v \hookrightarrow \text{Frac } \mathcal{O}_v = F_X$ , and the right map comes from the identification of  $X$  with the special fibre  $\mathcal{X}_s$ . Specifically, it is the canonical map  $\text{Spec } F_X \cong \text{Spec } \mathcal{O}_{\mathcal{X}, \xi} \rightarrow \mathcal{X}$ , where  $\xi$  is the generic point of  $\mathcal{X}$ .

**Lemma 1.21.** *There exists a unique centre  $x \in \mathcal{X}$  of  $v$ .*

*Proof.* By the valuative criterion for properness, such a centre exists, and the morphism  $f$  is unique (see also [?, Section 01J5, Lemma 01J6]):

$$\begin{array}{ccc} \text{Spec } F_X & \xrightarrow{\quad} & \mathcal{X} \\ \downarrow & \searrow f & \downarrow \\ \text{Spec } \mathcal{O}_v & \xrightarrow{\quad} & \text{Spec } \mathcal{O}_K. \end{array}$$

□

**Lemma 1.22** ([?, 3.11]). *We have  $v \in V(\mathcal{X})$  if and only if the centre  $x$  of  $v$  is of codimension one. In that case  $x$  is the generic point on the special fibre which induces  $v$ .*

*Proof.* Assume  $v \in V(\mathcal{X})$ . Then  $v$  is induced by a generic point  $\eta \in \mathcal{X}_s$  on the special fibre. The canonical morphism  $\text{Spec } \mathcal{O}_{\mathcal{X},\eta} \rightarrow \mathcal{X}$  shows that  $v$  has centre  $\eta$ , which is indeed a point of codimension one. Conversely, suppose  $v \in V_{\text{II}}(F_X)$  has center  $x \in \mathcal{X}$  of codimension one. The morphism  $\text{Spec } \mathcal{O}_v \rightarrow \mathcal{X}$  uniquely factors through  $\text{Spec } \mathcal{O}_v \rightarrow \text{Spec } \mathcal{O}_{\mathcal{X},x}$ , so  $\mathcal{O}_v$  dominates  $\mathcal{O}_{\mathcal{X},x}$ . Since these are both valuation rings with common fraction field  $K$ , it follows that they are in fact equal. Thus  $v \in V(\mathcal{X})$ .  $\square$

**Lemma 1.23** ([?, 3.12, 3.13]). *Let  $x \in \mathcal{X}$  be the centre of  $v$ , and let  $f \in \mathcal{O}_v^* \subset F_X$  be such that the reduction  $\bar{f} \in k(v)$  is transcendental over  $k$ . Then:*

- (a) *If  $f \in \mathcal{O}_{\mathcal{X},x}$ , then  $v \in V(\mathcal{X})$*
- (b) *If the rational map  $\mathcal{X} \dashrightarrow \mathbb{P}^1(\mathcal{O}_K)$  induced by  $f$  is defined at the center  $x$  of  $v$ , then  $v \in V(\mathcal{X})$ .*

*Proof.* (a) By Lemma 1.22, it suffices to show that  $x$  is a point of codimension one. Choose an open affine set  $\text{Spec } A = U \subset \mathcal{X}$  such that  $x \in U$  and  $f \in A$ . Denote by  $\mathfrak{p} \subset A$  the prime ideal corresponding to  $x$ . Since  $v$  is non-trivial, the center  $x$  is not the generic point of  $\mathcal{X}$ , and so the codimension is not zero. If the dimension of  $A/\mathfrak{p}$  is at least one, then the inequality

$$\dim(A_{\mathfrak{p}}) + \dim(A/\mathfrak{p}) \leq \dim(A) = 2$$

implies

$$\text{codim}(\overline{\{x\}}, \mathcal{X}) = \dim(A_{\mathfrak{p}}) \leq 2 - \dim(A/\mathfrak{p}) \leq 2 - 1 = 1,$$

and hence equals one. Thus, it suffices to show that  $A/\mathfrak{p}$  is not a field. If  $A/\mathfrak{p}$  were a field, it would be a finite extension of  $k$ , but this is impossible since  $\bar{f} \in A/\mathfrak{p} \subset k(v)$  is transcendental over  $k$ .

- (b) It suffices to show, using (a), that  $f \in \mathcal{O}_{\mathcal{X},x}$ . Let  $P = V(f^{-1})$  and  $Z = V(f)$  be the sets of poles and zeros of  $f$ , respectively. Then the rational map  $\mathcal{X} \dashrightarrow \mathbb{P}^1(\mathcal{O}_K)$  is obtained by gluing the morphisms

$$\mathcal{X} \setminus P \rightarrow \text{Spec } \mathcal{O}_K[t] \subset \mathbb{P}^1(\mathcal{O}_K), \quad \text{and} \quad \mathcal{X} \setminus Z \rightarrow \text{Spec } \mathcal{O}_K[t^{-1}] \subset \mathbb{P}^1(\mathcal{O}_K),$$

which are induced by

$$\begin{array}{ccc} \mathcal{O}_K[t] \rightarrow \Gamma(\mathcal{X} \setminus P, \mathcal{O}_{\mathcal{X}}), & \text{and} & \mathcal{O}_K[t^{-1}] \rightarrow \Gamma(\mathcal{X} \setminus Z, \mathcal{O}_{\mathcal{X}}), \\ t \mapsto f & & t^{-1} \mapsto f^{-1} \end{array}$$

and is defined on  $U := (\mathcal{X} \setminus P) \cup (\mathcal{X} \setminus Z) = \mathcal{X} \setminus (P \cap Z)$ . By assumption,  $x \in U$ . If  $x \in \mathcal{X} \setminus P$ , the claim follows immediately, as

$$f \in \Gamma(\mathcal{X} \setminus P, \mathcal{O}_{\mathcal{X}}) \subset \mathcal{O}_{\mathcal{X},x}.$$

If  $x \in \mathcal{X} \setminus Z$ , then

$$f^{-1} \in \Gamma(\mathcal{X} \setminus Z, \mathcal{O}_{\mathcal{X}}) \subset \mathcal{O}_{\mathcal{X},x}.$$

The image of  $f^{-1}$  under the local ring homomorphism  $\mathcal{O}_{\mathcal{X},x} \rightarrow \mathcal{O}_v$  is a unit by assumption, hence  $f^{-1}$  is also a unit in  $\mathcal{O}_{\mathcal{X},x}$ . Consequently  $f \in \mathcal{O}_{\mathcal{X},x}$ . □

We can now finish the proof of Proposition 1.19 i.

*Proof of Proposition 1.19 i.* Let  $f \in \mathcal{O}_v^\times \subset F_X$  be such that the reduction  $\bar{f} \in k(v)$  is transcendental over  $k$ . We also write  $f: \mathcal{X} \dashrightarrow \mathbb{P}^1(\mathcal{O}_K)$  for the rational map induced by  $f$ , and let  $U$  be its domain of definition. Define  $\Gamma_f \subset \mathcal{X} \times_{\mathcal{O}_K} \mathbb{P}^1(\mathcal{O}_K)$  to be the graph of  $f$ . Recall that it is defined to be the closure of the graph  $f|_U$  in  $\mathcal{X} \times_{\mathcal{O}_K} \mathbb{P}^1(\mathcal{O}_K) \supset U \times_{\mathcal{O}_K} \mathbb{P}^1(\mathcal{O}_K)$ . This is almost a model of  $X$ , as it satisfies all the properties of a model except that it might not be normal.

$$\begin{array}{ccccc} & & \mathcal{X}' & & \\ & & \downarrow & & \\ \mathcal{X} \times_{\mathcal{O}_K} \mathbb{P}^1(\mathcal{O}_K) & \supseteq & \Gamma_f & \longrightarrow & \mathbb{P}^1(\mathcal{O}_K) \\ & & \downarrow & \nearrow f & \\ & & \mathcal{X} & & \end{array}$$

The first projection  $\mathcal{X} \times_{\mathcal{O}_K} \mathbb{P}^1(\mathcal{O}_K) \rightarrow \mathcal{X}$  induces a birational morphism  $\Gamma_f \rightarrow \mathcal{X}$ . The second projection induces a morphism  $\Gamma_f \rightarrow \mathbb{P}^1(\mathcal{O}_K)$  which extends  $f: \Gamma_f \dashrightarrow \mathbb{P}^1(\mathcal{O}_K)$ . Let  $\mathcal{X}' \rightarrow \Gamma_f$  denote the normalization of  $\Gamma_f$ . This gives a model of  $\mathcal{X}'$  for which  $f: \mathcal{X}' \rightarrow \mathbb{P}^1(\mathcal{O}_K)$  is a morphism. By Lemma 1.23 b, this implies that  $v \in V(\mathcal{X}')$ . □

## 1.2.2 Proof of the Second Part

We fix a model  $\mathcal{X}$  of  $X$  and a nonempty subset  $V \subset V(\mathcal{X})$ . Denote by  $\mathcal{E}$  the set of irreducible (integral) components of  $\mathcal{X}_s$  that induce the valuations in  $V(\mathcal{X}) \setminus V$ .

**Definition 1.24.** A model  $\mathcal{X}'$  together with a modification  $\varphi: \mathcal{X} \rightarrow \mathcal{X}'$  such that for every integral vertical curve  $E$  on  $\mathcal{X}$ , the set  $\varphi(E)$  is a point if and only if  $E \in \mathcal{E}$  is called a *contraction* of the  $E \in \mathcal{E}$ .

It can be shown that if a contraction exists, it is unique up to unique isomorphism. Our goal now is to show that a contraction of  $\mathcal{E}$  exists. Once this is done, the proof is complete, since for the contraction  $\varphi: \mathcal{X} \rightarrow \mathcal{X}'$ , we have by construction  $V(\mathcal{X}') = V$ .

**Lemma 1.25** ([?, Lemma 8.3.29, Lemma 8.3.3]). *Let  $\mathcal{L}$  be an invertible sheaf on  $\mathcal{X}$  generated by global sections  $s_0, \dots, s_n$ . Consider the morphism  $f: \mathcal{X} \rightarrow \mathbb{P}^n(\mathcal{O}_K)$  associated to these sections. Let  $E$  be a vertical curve of  $\mathcal{X}_s$  with the reduced induced closed subscheme structure. Then  $f(E)$  is reduced to a point if and only if  $\mathcal{L}|_E \cong \mathcal{O}_E$ .*

**Proposition 1.26** ([?, Proposition 8.3.30]). *The following are equivalent.*

- (a) *The contraction  $\varphi: \mathcal{X} \rightarrow \mathcal{X}'$  of  $\mathcal{E}$  exists.*
- (b) *There exists a Cartier divisor  $D$  on  $\mathcal{X}$  such that  $\deg(D|_{\mathcal{X}_g}) > 0$ , that  $\mathcal{L} \cong \mathcal{O}_{\mathcal{X}}(D)$  is generated by its global sections, and that for any integral vertical curve  $E$ , we have  $\mathcal{L}|_E \simeq \mathcal{O}_E$  if and only if  $E \in \mathcal{E}$ .*

*Proof.* We only sketch (b)  $\implies$  (a). For the other implication and details see [?, Proposition 8.3.30].

As  $\mathcal{X}_g$  is an integral projective curve over a field,  $\mathcal{L}|_{\mathcal{X}_g}$  is an ample divisor ([?, Proposition 7.5.5]). Replacing  $D$  by a positive multiple (which does not change the set  $\mathcal{E}$ ), we can suppose that  $\mathcal{L}|_{\mathcal{X}_g}$  is very ample.

Let  $f: \mathcal{X} \rightarrow \mathbb{P}^n(\mathcal{O}_K)$  be a morphism associated to sections  $s_0, \dots, s_n \in \Gamma(\mathcal{X}, \mathcal{L})$  which generate  $\mathcal{L}$ . Let  $Z$  be the closed subset  $f(\mathcal{X}) \subset \mathbb{P}^n(\mathcal{O}_K)$  endowed with the reduced (hence integral) scheme structure. Then  $f$  induces a dominant morphism  $\mathcal{X} \rightarrow Z$  that we still denote by  $f$ . This is an isomorphism on the generic fibre because  $\mathcal{L}|_{\mathcal{X}_g}$  is very ample. In particular,  $f: \mathcal{X} \rightarrow Z$  is birational. Denote by  $\pi: \mathcal{X}' \rightarrow Z$  the normalization morphism. As  $\mathcal{X}$  is normal,  $f$  uniquely factors into  $\varphi: \mathcal{X} \rightarrow \mathcal{X}'$  followed by  $\pi: \mathcal{X}' \rightarrow Z$ .

$$\begin{array}{ccc} & & \mathcal{X}' \\ & \nearrow \varphi & \downarrow \pi \\ \mathcal{X} & \xrightarrow{f} & Z \end{array}$$

Let  $E$  be an integral vertical curve on  $\mathcal{X}$ . Then we have

$$f(E) \text{ is a point} \iff \mathcal{L}|_E \simeq \mathcal{O}_E \iff E \in \mathcal{E}$$

where the first equivalence is a consequence of Lemma 1.25 and the second one is true by assumption. We claim that  $\pi: \mathcal{X}' \rightarrow Z$  is a finite morphism. This then completes the proof because  $f(E)$  is reduced to a point if and only if  $\varphi(E)$  is a point. The morphism  $f$  is projective ([?, Corollary 3.3.32(e)]), so  $\mathbb{A} \cong f_*\mathcal{O}_{\mathcal{X}}$  is a coherent sheaf on  $Z$  ([?, Corollary 5.3.5]). Furthermore,  $\mathbb{A}$  is a sheaf of integrally closed algebras because  $\mathcal{X}$  is normal. As a result, the normalization morphism  $\pi: \mathcal{X}' \rightarrow Z$  coincides with the canonical morphism  $\text{Spec } \mathbb{A} \rightarrow Z$  and is a finite morphism ([?, Exercise 5.1.17]).  $\square$

For the construction of a suitable Cartier divisor, we furthermore need the following technical lemma.

**Lemma 1.27** ([?, Lemma 8.3.32]). *Let  $D$  be an effective horizontal Cartier divisor on  $\mathcal{X}$ , and  $\mathcal{L} = \mathcal{O}_{\mathcal{X}}(D)$ . Then there exists a  $m_0 \geq 1$  such that  $\mathcal{L}^{\otimes m}$  is generated by its global sections for every  $m \geq m_0$ .*

Proposition 1.26 and Lemma 1.27 show that for a contraction morphism to exist, it suffices that there exists an effective Cartier divisor that meets the closed fibres in a suitable way. To construct such a divisor, it is important that the ring  $\mathcal{O}_K$  is complete, which implies it is Henselian.

**Definition 1.28.** Let  $A$  be a Noetherian local ring with maximal ideal  $\mathfrak{m}$  and residue field  $k := A/\mathfrak{m}$ . We say that  $A$  is Henselian if one of the following equivalent conditions is fulfilled:

- Hensel's Lemma holds: For every monic  $f \in A[T]$ , and every simple root  $a_0 \in k$  of  $\bar{f} \in k[T]$ , there exists a lift  $a \in A$  (so  $\bar{a} = a_0$ ) such that  $f(a) = 0$ .
- Every finite  $A$ -algebra is a direct sum of local  $A$ -algebras.

**Lemma 1.29** ([?, 8.3.35]). *Let  $x \in \mathcal{X}_s$  be a closed point on the special fiber, then the following assertions hold.*

- (a) *There exists an irreducible horizontal Weil divisor  $\Delta$  containing  $x$ .*
- (b) *There exists an effective horizontal Cartier divisor  $D$  such that  $\mathcal{X}_s \cap D = \{x\}$ .*

*Proof.* (a) Let  $\mathfrak{m}_x$  be the maximal ideal of  $\mathcal{O}_{\mathcal{X},x}$  and denote by  $\pi$  an uniformizer of  $\mathcal{O}_K$ . Let  $\mathfrak{p}_1, \dots, \mathfrak{p}_n$  be the prime ideals of  $\mathcal{O}_{\mathcal{X},x}$  that are minimal among those containing  $\pi\mathcal{O}_{\mathcal{X},x}$ . As  $\dim \mathcal{O}_{\mathcal{X},x} = 2$  ([?, Proposition 8.3.4]), Krull's principal ideal theorem implies that  $\mathfrak{m}_x \neq \mathfrak{p}_i$ . There therefore exists an

$$f \in \mathfrak{m}_x \setminus (\cup_i \mathfrak{p}_i).$$

Thus  $f$  is a regular element and  $V(f)$  does not contain any irreducible component of  $\text{Spec } \mathcal{O}_{\mathcal{X},x}$ .

There exists an affine open subset  $W \ni x$  and a regular element  $g \in \mathcal{O}_{\mathcal{X}}(W)$  such that  $g_x = f$ , and that  $V(g) \cap W_s = \{x\}$ . Let  $\Delta$  be the closure in  $\mathcal{X}$  of an irreducible component of  $V(g)$  passing through  $x$ . Then  $\Delta$  is an irreducible horizontal divisor passing through  $x$ .

- (b) Let  $E := V(g)$  and let  $W$  be as in a. Then  $E$  can be seen as an effective Cartier divisor on  $W$ . Moreover,  $E \rightarrow \text{Spec } \mathcal{O}_K$  is quasi-projective and quasi-finite because  $\dim E_s < \dim \mathcal{X}_s = 1$ . By a corollary of Zariski's Main theorem ([?, Corollary 4.4.8]), there exists an affine open neighborhood  $U$  of  $x$  such that  $E \cap U$  is an open subscheme of a scheme  $Z$  that is finite over  $\text{Spec } \mathcal{O}_K$ .

$$E \cap U \xrightarrow{\text{open emb.}} Z \xrightarrow{\text{finite}} \operatorname{Spec} \mathcal{O}_K$$

Let  $D$  be the connected component of  $Z$  containing  $x$ . Since  $\mathcal{O}_K$  is a Henselian local ring,  $Z$  is a disjoint union of local schemes. Hence  $D$  is local, and consequently  $D \subset E \cap U \subset \mathcal{X}$ .

Finally, since  $D$  is closed in  $Z$ , it is finite over  $\operatorname{Spec} \mathcal{O}_K$ . A finite morphism  $D \rightarrow \operatorname{Spec} \mathcal{O}_K$  between affine schemes is projective ([?, Exercise 3.3.22]), and therefore proper ([?, Theorem 3.3.30]). So the morphisms

$$D \rightarrow \mathcal{X} \rightarrow \operatorname{Spec} \mathcal{O}_K \quad \text{and} \quad \mathcal{X} \rightarrow \operatorname{Spec} \mathcal{O}_K$$

are proper, which implies that  $D \rightarrow \mathcal{X}$  is also proper ([?, Proposition 3.3.16(e)]). Consequently,  $D$  is closed in  $\mathcal{X}$  and therefore a Cartier divisor, as required.  $\square$

Now we can finally prove Proposition 1.19 ii.

*of Proposition 1.19 ii.* Let  $Z_1, \dots, Z_n$  be the irreducible components of  $\mathcal{X}_s$  that induce the valuations in  $V \subset V(\mathcal{X})$ . By Lemma 1.29 b, there exists an effective horizontal Cartier divisor  $D_i$  such that  $\mathcal{X}_s \cap D_i$  is a point of  $Z_i$  that belongs to none of the components of  $\cdot$ . Let  $D = \sum_{i=1}^n D_i$ . By virtue of Lemma 1.27, if necessary replacing  $D$  by a multiple, we can even suppose that  $\mathcal{L} := \mathcal{O}_{\mathcal{X}}(D)$  is generated by its global sections.

The theorem then results from Proposition 1.26. Indeed,  $\mathcal{L}|_E \simeq \mathcal{O}_E$  if  $E \in \text{since } E \cap D = \emptyset$ ; if  $E \notin$ , then  $E$  is some  $Z_i$  and  $\deg \mathcal{L}|_E \geq \deg \mathcal{O}_{\mathcal{X}}(D_i)|_{Z_i} > 0$ , and hence  $\mathcal{L}|_E \not\simeq \mathcal{O}_E$ .  $\square$

*Remark 1.30.* The proof of Proposition 1.19 ii shows that actually all models  $\mathcal{X}$  of  $X$  are projective. Indeed one just has to choose  $V = V(\mathcal{X})$  in the proof to obtain an effective Cartier divisor  $D$  which meets all the irreducible components of the fiber  $\mathcal{X}_s$ . Since  $\mathcal{X}_s$  is a projective curve ([?, Exercise 7.5.4]),  $\mathcal{O}_{\mathcal{X}}(D)|_{\mathcal{X}_s}$  is ample ([?, Proposition 7.5.5]). It follows from [?, Corollary 5.3.24] that  $\mathcal{O}_{\mathcal{X}}(D)$  is ample. So  $\mathcal{X} \rightarrow \operatorname{Spec} \mathcal{O}_K$  is projective ([?, Proposition 5.3.6]).

## 1.3 Permanent Models

### 1.3.1 Models with Reduced Special Fibre

As in the last two sections, we assume  $X$  to be a smooth, projective and absolutely irreducible curve over the local field  $K$ .

**Definition 1.31.** A model  $\mathcal{X}$  of  $X$  is called *permanent*, if  $\mathcal{X} \times_{\mathcal{O}_K} \mathcal{O}_L$  is normal for all finite field extensions  $L|K$ .

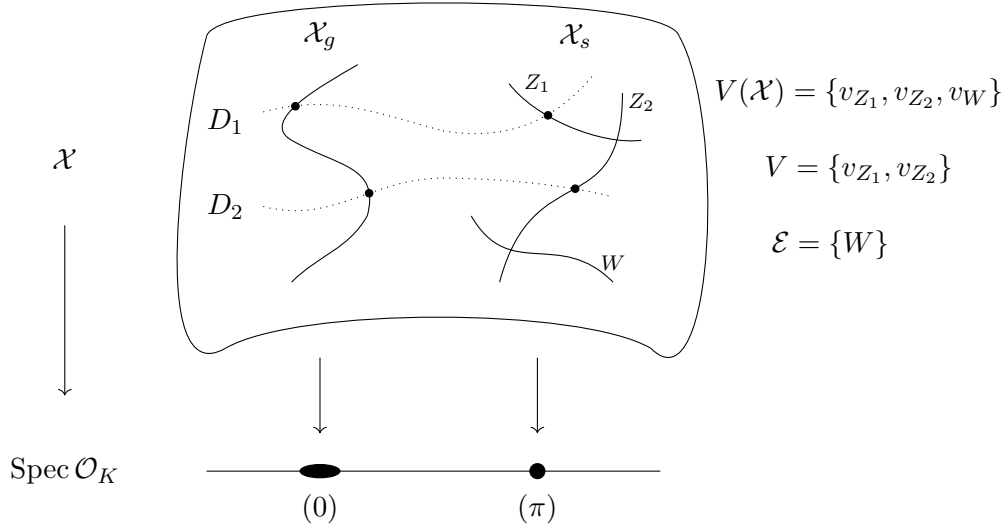


Figure 1.1: Illustration of the proof of Proposition 1.19 ii

*Remark 1.32.* Since flatness and properness are preserved under base change, a model  $\mathcal{X}$  of  $X$  is permanent if and only if  $\mathcal{X} \times_{\mathcal{O}_K} \mathcal{O}_L$  is a model of  $X_L = X \times_K L$ .

**Definition 1.33.** A *pre-model* of  $X$  is a flat and proper  $\mathcal{O}_K$ -scheme  $\mathcal{X}$  with  $\mathcal{X} \times_{\mathcal{O}_K} K = X$ . That is,  $\mathcal{X}$  satisfies all the conditions of a model except of being normal.

**Lemma 1.34** ([?, Lemma 1.3]). *Let  $\mathcal{X}$  be a pre-model of  $X$ . Then  $\mathcal{X}$  is normal (i.e. a model) if and only if the following two conditions hold.*

- (i) *The special fibre  $\mathcal{X}_s$  has no embedded points, i.e. no point  $x \in \mathcal{X}_s$  satisfying  $\mathfrak{m}_x = \text{Ann}_R(r)$  for some  $r \in R := \mathcal{O}_{\mathcal{X}_s, x}$  (see [?, Definition 7.1.6]).*
- (ii) *The local ring  $\mathcal{O}_{\mathcal{X}_\eta}$  is a discrete valuation ring for every generic point  $\eta \in \mathcal{X}_s$ .*

*Proof.* According to Serre's criterion for normality [?, Theorem 8.2.23], we need to check the following properties:

(R<sub>1</sub>)  $\mathcal{X}$  is regular at  $x$  when  $\text{codim } x \leq 1$ , and

(S<sub>2</sub>)  $\text{depth } \mathcal{O}_{\mathcal{X}, x} \geq 2$ .

for all  $x \in \mathcal{X}$ .

The points on  $\mathcal{X}$  with codimension 1 are precisely the points on the generic fibre, which is regular by assumption, and the generic point on the special fibre. Since any discrete valuation ring is a local Noetherian domain and vice

versa,  $(R_1)$  is equivalent to (i). By [?, Proposition 8.2.11], condition  $(S_2)$  is equivalent to say that  $\text{depth } \mathcal{O}_{\mathcal{X}_s, x} \geq 1$  for all  $x \in \mathcal{X}_s$ . But this is always true for the generic point, and if  $x$  is a different point it is equivalent to say that  $x$  is not an embedded point.  $\square$

**Example 1.35.** 1. Consider a pre-model with special fibre

$$\mathcal{X}_s := \text{Spec } k[x, y]/(x^2, xy), \quad (1.9)$$

which is topologically equivalent to  $\mathbb{A}^1(k)$ . The point corresponding to the maximal ideal  $(x, y)$  is an embedded point.

2. [?, Example 9.8.4].

**Proposition 1.36** ([?, Proposition 2.3]). *If  $\mathcal{X}_s$  is reduced, then  $\mathcal{X}$  is permanent.*

*Proof.* Let  $L|K$  be a finite extension and  $l|k$  the extension of the corresponding residue fields. The special fibre of  $\mathcal{X}' := \mathcal{X}_{\mathcal{O}_L} = \mathcal{X} \times_{\mathcal{O}_K} \mathcal{O}_L$  is  $\mathcal{X}'_s := \mathcal{X}_s \times_k l$ , which is reduced since  $k$  is perfect by assumption. This means  $\mathcal{X}'_s$  satisfies condition (ii) from Lemma 1.34.

For condition (i) let  $\eta \in \mathcal{X}'_s$  be any generic point. The chalk  $\mathcal{O}_{\mathcal{X}', \eta}$  then is a Noetherian local domain of dimension 1 and it holds

$$\mathcal{O}_{\mathcal{X}'_s, \eta} = \mathcal{O}_{\mathcal{X}', \eta} / \pi_L \mathcal{O}_{\mathcal{X}', \eta}, \quad (1.10)$$

where  $\pi_L \in \mathcal{O}_L$  is a uniformising parameter. Since  $\mathcal{X}'_s$  is reduced,  $\pi_L \mathcal{O}_{\mathcal{X}', \eta}$  is maximal ideal which means that condition (i) from Lemma 1.34 is satisfied as well. Thus  $\mathcal{X}'$  is normal and hence  $\mathcal{X}$  is permanent.  $\square$

**Proposition 1.37** ([?, Corollary 2.2 (ii)]). *Let  $\mathcal{X}$  be a model of  $X$ , then the following statements are equivalent.*

- (i)  $\mathcal{X}_s$  is reduced.
- (ii)  $\mathcal{O}_{\mathcal{X}_s, \eta}$  is a field for every generic point  $\eta \in \mathcal{X}_s$ .
- (iii) Every  $v_Z \in V(\mathcal{X})$  is weakly ramified.

*Proof.* Since  $\mathcal{X}$  is normal, there are no embedded points in  $\mathcal{X}_s$  by Lemma 1.34.  $\square$